THE ETALE LOCAL RING IS THE STRICT HENSELIZATION OF THE HENSELIAN LOCAL RING

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 2772-2781, immediately following the statement at 2761-2770.

Editorial note. The equality (R^h)^sh = R^sh holds for the strict henselization taken with the separable closure kappa^sep fixed by the geometric point; the reviewer should read both descriptions against that fixed choice.

Editorial note. Two cross-chapter pointers (Properties of Spaces, Algebra) and one same-chapter stub (lemma-describe-henselian-local-ring, carded as L26052) are recorded in assembly.external_references.

Allowed standard prerequisites (editorial).

- Etale local ring described via charts. Properties of Spaces, Lemma spaces-properties-lemma-describe-etale-local-ring, cited at proof lines 2774-2776. Identifies O_{X,xbar} with the strict henselization of the chart local ring O_{U,u}.
- Henselian local ring via elementary etale neighbourhoods. Lemma lemma-describe-henselian-local-ring of this chapter, cited at proof line 2773. O^h_{X,x} = O^h_{U,u} – the bridge pairing with the previous item.
- Strict henselization of a henselian ring. Algebra, Lemma algebra-lemma-uniqueness-henselian, cited at proof lines 2779-2781. (R^h)^sh = R^sh for a local ring with a chosen separable closure of the residue field.

Source reference. Original reference 08HT: focused-seeds/geometry/S9-decent-henselian-local-ring-strict/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0EMY: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 2732–2732. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04KG: focused-seeds/geometry/S9-decent-henselian-local-ring-strict/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:decent-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 04KF: focused-seeds/geometry/S9-decent-henselian-local-ring-strict/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let X be a decent algebraic space over S. Let $\overline{x}$ be a geometric point of X lying over $x \in |X|$. The étale local ring $\mathcal{O}_{X,\overline{x}}$ of X at $\overline{x}$ (Properties of Spaces, Definition 04KG) is the strict henselization of the henselian local ring $\mathcal{O}^h_{X,x}$ of X at x.*

Proof. Follows from Lemma 0EMY, Properties of Spaces, Lemma 04KF and the fact that $(R^h)^{sh} = R^{sh}$ for a local ring $(R, \mathfrak{m}, \kappa)$ and a given separable algebraic closure κ^{sep} of κ. This equality follows from Algebra, Lemma 08HT. □